System Architecture Overview

GlucoTwin: Multimodal Digital Twin System Architecture

1. Data Ingestion

Static EHR Stream

- Patient Demographics
- Baseline HbA1c
- BMI & Vitals
- Clinical Regimen

Dynamic Sensor Stream

- 5-Min CGM Telemetry
- Continuous HRV
- Step & Activity Logs

DPDP Act 2023 & HIPAA Compliant Sandbox

2. Feature Encoding Layer

Dense EHR Embedding Layer

Categorical & numerical projection

`[32D Vector]`

2-Layer LSTM Temporal Encoder

Sequential time-series encoding

`[64D Vector]`

3. Fusion & Risk Engine

Late Fusion Network

Cross-attention concatenation

`[96D Joint Latent Space]`

2-Hour Forward Horizon Predictor

Continuous Trajectory Forecast

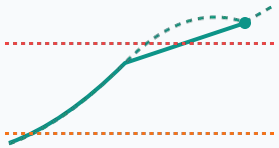
⚠ Hyperglycemia Alert
Threshold: > 180 mg/dL
Triggers insulin prompt

⚠ Hypoglycemia Alert
Threshold: < 70 mg/dL
Triggers carb intake prompt

4. Clinical Delivery

Doctor Streamlit Cockpit

Live Trajectory Visualizer



What-If Simulator
Physiological Counterfactuals

Meal Intake Model
Simulate +45g Carbs

Exercise Model
Simulate 30-Min Walk